



Additional Safety Measures for the 793B

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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1.0 Introduction

Installing aftermarket electrical devices (e.g. warning systems) on a 793B by modifying the machine's electrical harnesses can generate faults in the electrical system due to loose contact, overloaded circuit system, etc.

This Best Practice is achieved by using CAT parts without making modification to the system.

2.0 Best Practice Description

Installing third party accessories may cause a delay in the commissioning of the 793B. Electrical faults created by third party devices may also cause lower machine availability due to lack of information and knowledge of the original circuitry. This Best Practice is to have the work performed by a CAT dealer using original Cat parts.

3.0 Implementation Steps

There have been complaints of electrical faults generated after installing third party devices after the delivery of the 793B to customer's mine site. Electrical faults were generated by third party technicians handling the 793B circuits. A list of additional accessories installed by the customer:

- Siren boot. (sounds during startup)
- Alert sound inside the cabin when the machine is stopped (for 20 sec.) without parking
- External visual alert when the machine is operating but with the parking brake applied
- Passing light (used to indicate to other vehicles that can overtake the truck)

With this information a schematic was drawn with Cat parts.

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4.0 Benefits

- Visual alert ensures that the truck is parked.
- The truck passing light is used to authorize a light vehicle to overtake the truck.
- Can be installed without changing the wiring, using original spare Cat parts.
- Improves electrical monitoring to maintain the original wiring.
- Improved availability in the early hours compared to a third party.
- Improves the truck delivery time, when installed during repair.

5.0 Resources Required

- For this Best Practice, Caterpillar technical information coupled with the experience of the technical staff with support from supervisor of the workshop was used.
- 60 hours of work.

6.0 Supporting Attachments / References

793B IMPROVEMENT		
BLOCK FUSE		
CANT	N° PART	DESCRIPTION
1	1747566	Block as fuse
16	9w2033	Terminal
4	6v5645	Screw
6	9w1441	Fuse de 15 a
6	9w1442	Fuse de 10a
4	9w1446	Fuse de 20 ^a
2	1028806	Receptacle
2	1552265	Plug
1	2341193	Wire as

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IMPROVEMENT PASSING LIGHT		
CANT	N° PART	DESCRIPTION
2	321-7871	Lamp
6	5C2890	Nut
6	9X8256	Washer
1	7N-8001	Switch
120	3668856	Dm Harness
1	1330970	Receptacle
3	1552260	Receptacle
1	1305300	Clip
50	7K1181	Tie
1	9X1054	Plug relay
6	9x1056	Terminal
1	3E5239	Relay
20	9w0844	Socket
20	9w0852	Pin
20	2L8076	Terminal
2	5M2894	Washer
2	4B-2049	NUT
2	6D-6146	SCREW
2	0S1594	BOLT

IMPROVEMENT BACK UP LIGHT PARKING		
CANT	N° PART	DESCRIPTION
100	3668856	Dm Harness
1	9X1054	Plug
6	9x1056	Terminal
3	1552270	Plug
3	1028802	Receptacle
10	9w0844	Socket
10	9w0852	Pin
1	3E-2033	Pressure Switch
50	3s2093	Tie
1	3E5239	Relay
50	1257875	Tube

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IMPROVEMENT PARKING ALARM		
CANT	N° PART	DESCRIPTION
50	3668856	Dm Harness
1	9X1054	Plug relay
1	3E5239	Relay
6	9X1056	Terminal
1	9G9813	Alarm
50	1193662	Cm tube
20	9W0844	Socket
20	9W0852	Pin
1	3E2034	Switch de presión
50	3S2093	Tie
2	2D7325	Tee
1	1893158	Diodo
4	1552270	Plug
4	1028802	Receptacle
1	4W8471	Relay
2	9W3087	Terminal
1	2L8079	Terminal
6	5P1476	Terminal
2	1364877	Splice wire
10	6V8837	Bolt
8	9X8256	Washer

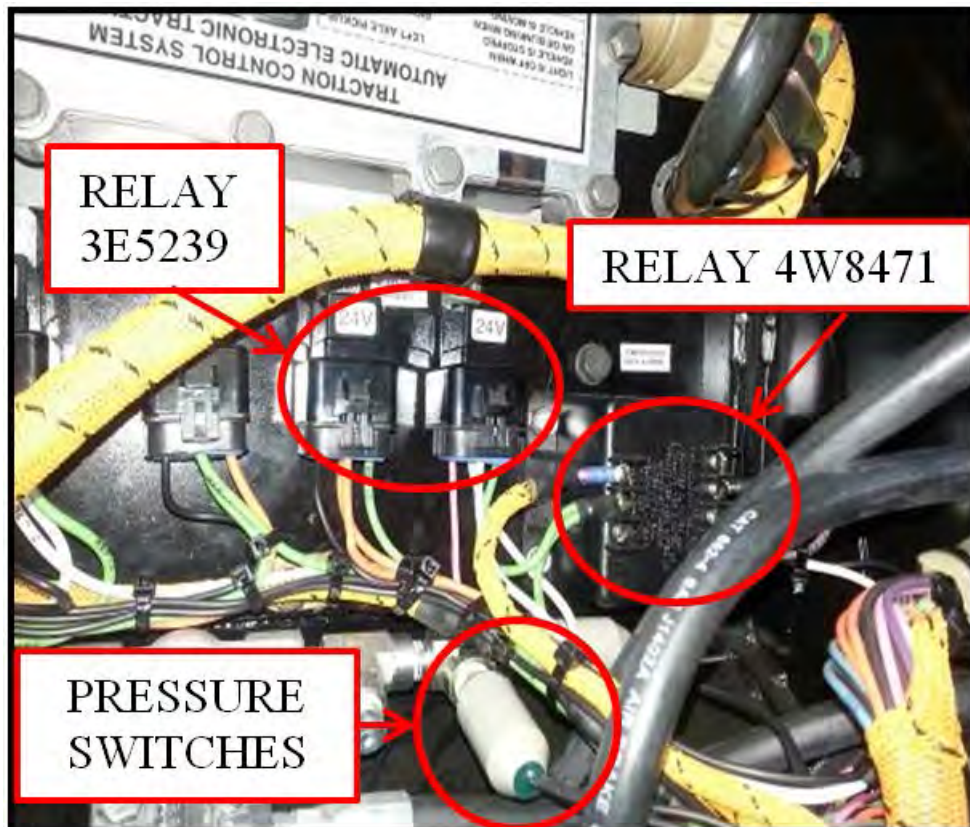
IMPROVEMENT START ALARM		
CANT	N° PART	DESCRIPTION
1	8D-2676	Switch
120	3668856	Dm Harness
1	1330970	Receptacle
3	1552260	Receptacle
1	1305300	Clip
50	7K1181	Tie
1	9X1054	Plug relay
6	9x1056	Terminal
1	3E5239	Relay
50	9w0844	Socket
50	9w0852	Pin
10	2L8076	Terminal

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CAB REAR.



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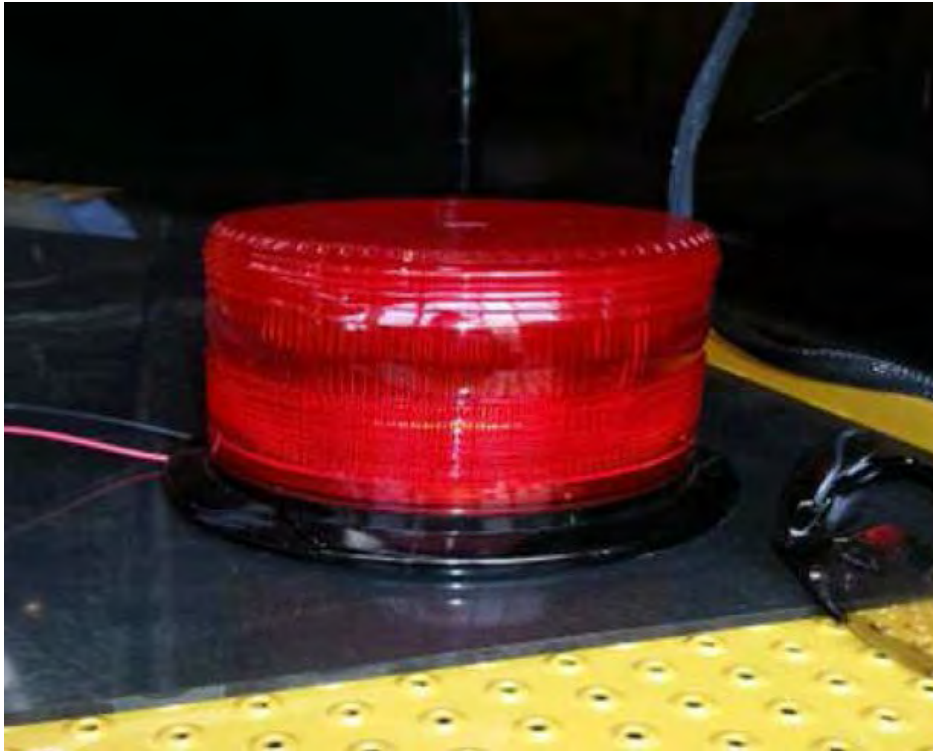
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Back up light parking recommended.



REAR MACHINE



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7.0 Related Best Practices

None.

8.0 Acknowledgements

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